CCC-C-425D
December 13, 1971
SUPERSEDING
Fed. Spec. CCC-C-425C
May 15, 1967

FEDERAL SPECIFICATION

CLOTH, BIRDSEYE AND GAUZE: COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of cotton cloth.

1.2 Classification.

1.2.1 Types and classes. The cloth shall be of the following types and classes as specified (see 6.2):

Type I - Birdseye.

Class 1 - Diaper cloth (bleached).
Class 2 - Coverall cloth (bleached).
Class 3 - Coverall cloth (dyed).

Type II - Gauze - double layer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

FSC 8305

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Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

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3.2 Material.

3.2.1 Cotton. The yarn shall be made from cotton and shall be carded, drawn, and spun.

3.3 Color.

3.3.1 Bleached. The color for type I, classes 1 and 2 and type II cloth shall be chemically bleached to match the available standard sample (see 6.3). An optical brightener or tinting compound may be used unless otherwise specified. The bleached finished cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.4. When no standard sample is available, the discoloration shall be no more than the rating "fair" when tested as specified in 4.4.

3.3.2 Dyed. The color for type I, class 3 cloth shall be OG-107 or as specified (see 6.2 and 6.4). The use of dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyed and finished cloth shall contain no more labile sulfur than that shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.3.3 Matching. The color shall match the available standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.4 Colorfastness. The dyed cloth shall show colorfastness to combined laundering and bleaching, perspiration, crocking, and light equal to or better than the standard sample. When no standard sample is established or when the standard sample is not a reference for colorfastness, the dyed cloth shall show "good" colorfastness to combined laundering and bleaching, perspiration, and light, and no more crocking than Munsell Value 8.5 when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I or table II as applicable when tested as specified in 4.4.

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TABLE I. Physical requirements.- type I, birdseye

Class	Weight per square yard (ounces)		Width (inches (including selvage)		Yarns per inch (minimum)		Breaking strength (pounds) minimum	
	Min.	Max.	Min.	Max.	Warp	Filling	Warp	Filling
1	4.0	-----	26-1/2	27-1/2	60	48	40	60
2	3.4	3.8	as specified		60	46	40	60
3	3.7	4.2	as specified		60	46	40	60

TABLE II. Physical requirements.- type II, gauze

Weight 1/ (oz/sq.yd.)	Width (including selvages) inches		Yarns per inch (sum of warp and filling) 2/	Breaking strength 1/ (pounds) minimum	
Minimum	Min.	Max.	Minimum	Warp	Filling
3.2	39-1/2	40-1/2	74	45	25

1/ Utilizing two plys of panel.

2/ Determined on single ply of panel, but both single panels shall conform to this requirement.

3.4.1 Weave.

3.4.1.1 Type I, classes 1, 2, and 3. The cloth shall be a single layer cloth in a birdseye weave.

3.4.1.2 Type II. Type II cloth shall be a double layer gauze cloth woven in a plain interlocking-type weave. The combining, or interlocking, of the double layer to form a single layer shall be throughout the length of the cloth and at regular intervals filling-wise to form pinking bars.

3.4.1.2.1 Pinking bar. A filling-wise bar (pinking bar) 1-1/2 ± 1/2 inches wide shall be woven into the cloth at regular intervals so as to form panels of the double layer (2-ply cloth). The 1-1/2 inch pinking bars shall be made by weaving the warps of the 2-ply cloth to form a single layer cloth.

3.4.1.2.2 Panels. The panels of 2-ply cloth formed by the pinking bars shall measure 19 ± 1/2 inch in length from inside of pinking bar to inside of pinking bar.

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3.4.1.2.3 Folding stripes. The supplier may, at his option, insert folding stripes. If folding stripes are used, one stripe consisting of four ends weaving as one in a group of 3 or 4 shall be positioned 13-1/2 ± 1/2 inch from one selvage. Another stripe shall be positioned 8 ± 1/2 inch from the opposite selvage. The folding stripes shall become part of the single layer pinking bars at point of intersection.

3.5 Shrinkage. The sum of the warp and filling residual shrinkage for type I, class 1, or type II cloth shall not exceed 20 percent when tested as specified in 4.4. The residual shrinkage for type I, classes 2 and 3 cloth shall not exceed 2.0 percent in either the warp or filling directions when tested as specified in 4.4.

3.6 Absorbency. Type I, class 1 and type II cloth shall show a minimum absorption of 80 percent when tested as specified in 4.4. Type I, classes 2 and 3 cloth shall show a minimum absorption of 65 percent when tested as specified in 4.4.

3.7 Finish. No additive finishes shall be applied and the character of the finished cloth shall be equal to that of the standard sample.

3.7.1 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.4 except that the determination for type I, class 3 shall be prior to dyeing.

3.8 pH.- The pH value of the water extract of the finished cloth shall be not less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (see 6.6). The cloth shall be put up on full width rolls as specified in PPP-P-1134.

3.9.1 Pieces. Pieces of type II cloth shall begin and end with a pinked edge located in the middle of the pinking bar.

3.10 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.11 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptance quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.2.2 Examination of the end item. Defects found during this examination shall be classified in accordance with 4.2.2.1, through 4.2.3.2.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be 1 yard, full width of the finished cloth. The acceptable quality level shall be 6.5 defects per 100 units (yards). The sample size shall be in accordance with inspection level 1 of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with table III.

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Defects

Hole, cut or tear
Smash
Knot 1/
Grease or oil stain
Embedded crease or wrinkle

1/ To be scored when larger than limit shown for knots on figure 1 of Fed. Std. No. 4.

4.2.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table III.

Defects

Selvage curled, folded, rolled, or slack continuously or intermittently throughout the piece.
Over-all uncleanliness. 1/
Objectionable odor.
Uneven weaving throughout the roll.
Not bleached white (type I, classes 1 and 2, and type II).
Not color as specified (type I, class 3).
Poor dye penetration, mottled, streaky, or cloudy (type I, class 3).
Width not within specified tolerance. 1/
Length of panels not within specified tolerance (type II).
Width of pinking bar not within tolerance (type II).
Folding stripes, if present, not in specified locations (type II).
Any piece which does not begin or end with a pinked edge in the middle of the pinking bar. (type II).
Not labeled in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

1/ Shall not be scored unless it occurs for more than 5 yards per roll.

TABLE III. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3200	5	0

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TABLE III. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	1
35,001 up to and including 150,000	20	2
150,001 and over	32	3

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.2.3 Examination for length.

4.2.3.1 Individual rolls. The rolls shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table III.

4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for examination of individual rolls.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine whether the packaging, packing, and marking complies with the section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table IV, shall be followed. Except as otherwise indicated, the physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3 continuous yards, full width, of the finished cloth and 1/4 yard full width of type I, class 3 cloth prior to dyeing. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

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Lot size (yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Test methods

Characteristic	Requirement paragraph	Test method
Material, cotton	3.2.1	1200 1/
Discoloration	3.3.1	5660 2/ 3/
Presence of labile sulfur	3.3.2	2020
Colorfastness to:		
Combined laundering and bleaching	3.3.4	5605
Perspiration	3.3.4	5680
Light	3.3.4	5660
Crocking	3.3.4	5651
Weight	3.4	5041 4/
Yarns per inch:		
Warp	3.4	5050 5/
Filling	3.4	5050 5/
Breaking strength:		
Warp	3.4	5100 6/
Filling	3.4	5100 6/
Weave	3.4.1	Visual 3/
Shrinkage in laundering:		
Warp	3.5	5550 7/
Filling	3.5	5550 7/
Absorbency	3.6	5502
Nonfibrous materials	3.7.1	2611
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the requirements specified.

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2/ Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration of the specimen shall be considered satisfactory if it is not worse than the rating of "fair" indicated in method 5660.

3/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

4/ For type II - determination to be made on the two layers of body cloth and should not include folding stripes if present.

5/ For type II - determination to be made on single layer of panel, specimens shall be sampled from both layers and must meet the requirement. The specimens should not include folding stripes if present.

6/ For type II - determination to be made of double layer body cloth (not single ply). All tests must be clear of, and must not include the pinking bars or folding stripes, if present.

7/ The specimens of type II cloth shall be a complete diaper, double layer, and shall be cut the width of the cloth through the middle of the respective pinking bars, leaving a pinked edge. The samples shall be dried prior to pressing.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Cloth of one type and class only shall be packed in accordance with the applicable requirements of PPP-P-1134.

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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloths covered by this specification are intended for the following:

(a) Type I, class 1 cloth and type II cloth are intended for use in the fabrication of infant diapers.
(b) Type I, classes 2 and 3 cloths are intended for use in the fabrication of coveralls, cooling, toxicological agents protective and rocket fuel handlers. Class 2 is used by the Air Force and class 3 is used by the Army.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of cloth required (see 1.2.1).
(c) Color of cloth required (see 3.3.2).
(d) Width required (see 3.4).
(e) Minimum length, if other than specified (see 3.9).
(f) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Dyestuff formulation.

6.4.1 Olive Green 107. A suggested but not mandatory dyestuff formulation for Olive Green 107 (OG 107) is as follows:

New color index

Vat Black 25, CI 69525

Vat Green 3, CI 69500

Vat Green 8, CI 71050

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Shaded with either or both of the following:

Vat Brown 3, CI 69015

Vat Yellow of suitable fastness

The redness of shade should result from the predominant use of the main colors specified and not from the shading colors.

6.5 For GSA procurement, rolls or bolts shall be 50 yards in length with a tolerance of within plus or minus 2-1/2 yards.

MILITARY CUSTODIANS:

Army - GL
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Air Force - 82

Review activities:

Army - MD
Navy - MS

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CIVIL AGENCY INTEREST:

COM-NBS
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Cloth, Birdseye And Gauze: Cotton CCC-C-425D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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